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Hybrid features and optimization-driven recurrent neural network for glaucoma detection

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Abstract

Glaucoma is considered as the main source of irrevocable loss of vision. The earlier diagnosis of glaucoma is essential to provide earlier treatment and to reduce vision loss. The fundus images are transfigured in the ophthalmology and are used to visualize the structures of the optic disc. However, accuracy is considered as a major constraint. To increase accuracy, an effective optimization-driven classifier is developed for glaucoma detection. The proposed Jaya-chicken swarm optimization (Jaya-CSO) is employed for training the recurrent neural network (RNN) for glaucoma detection. The proposed Jaya-CSO is designed by integrating the Jaya algorithm with the chicken swarm optimization (CSO) technique for tuning the weights of the RNN classifier. The method utilized optic disc features, statistical features, and blood vessel features for the determination of the glaucomatous region. The features obtained from the optic disc, blood vessels, and the fundus image is formulated as a feature vector. Finally, the glaucoma classification is done using RNN using the feature vector such that the RNN is trained using the proposed Jaya-CSO. The proposed Jaya-CSO outperformed other existing models with maximal accuracy of 0.97, the specificity of 0.97, and sensitivity of 0.97, respectively.

KEYWORDS

glaucoma detection, optic discs, Renyi entropy, segmentation, sparking process

Abbreviations: ACA, anterior chamber angle; ACG, angle-closure glaucoma; ANN, artificial neural network; AS-OCT, anterior segment-optical-coherence-tomography; CDR, cup-to-disc ratio; CSO, chicken swarm optimization; DCNNs, deep convolutional neural networks; DR, diabetic retinopathy; DWT, discrete wavelet transforms; EWT, empirical wavelet transforms; GIST, gastrointestinal stromal tumor; HDT, holoentropy-based decision tree; HRF, high-resolution fundus; HSF, hybrid structural feature-set; HTF, hybrid texture feature-set; ILM, inner limiting membrane; IOP, intraocular pressure; Jaya-CSO, Jaya-chicken swarm optimization; LGP, local gradient pattern; LS-SVM, least square-support vector machine; OCT, optical-coherence-tomography; PCA, principal component analysis; PHOG, pyramid histogram of oriented gradients; PR, pattern recognition; RBF, radial basis function; RNN, recurrent neural network; RPE, retinal pigment epithelium; SVM, support vector machine.

1 | INTRODUCTION

Glaucoma is a kind of chronic eye disease wherein a loss of vision may occur due to the growing of optic nerve, which may injure the astrocytes leading to elevated IOP.¹ This glaucoma tends to be a second major root of blindness and visual destruction with lots of glaucoma cases, which may increase tremendously.² The earlier finding of this disease may slow the disease progression and prevent the loss of vision.³ Due to the complex character of glaucoma disease pathology, its diagnosis depends on expert ophthalmologists for detecting glaucoma in its earlier stage, so that the vision of the patient is preserved.⁴

However, the detection of glaucoma is a lengthy process and needs experts and advanced tools for diagnosis.⁵

The progression in processing medical images has facilitated the design of different modalities of images using advanced systems for detecting glaucoma with different features and classification methods.⁶⁻¹² The glaucoma is categorized into three categories like developmental glaucoma, ACG, and open-angle glaucoma. The categories are further divided into primary and secondary divisions. From these categories, the ACG is considered as an asymptomatic and persistent, which tends to be more ubiquitous.⁵ The ACG caused more visual morbidity as compared to open-angle glaucoma. The conventional diagnosis methods for detecting glaucoma are omnipresent, costly, and lengthy process.^{13,14}

The techniques used to diagnose glaucoma can reduce the variability and increase the accuracy. The two foremost modalities of imaging employed in diagnosing glaucoma are Funduscopy and OCT. The Funduscopy is a technique that is simpler to use and provides an exaggerated eye-view. However, it can incarcerate the changes of structures on the outmost retina layer that materialize the untreatable and last stage. There is a huge requirement for capturing the changes of internal structures devised by glaucoma with respect to the initial and treatable stage. The variations in the structures of the retina layers occurred before Visual loss.¹⁵ The timely and earlier analysis of these variations prevents the loss of vision.¹⁶

The contemporary modalities of the image, like OCT pose the abilities to generate reproducible computations of certain topographic parameters in the layers of the retina. The function of glaucoma is in the form of optical biopsy,^{17,18} which facilitates imaging retina in the form of 3D that could enable a detailed analysis of structural changes.¹⁹ The cup consists of an inner white piece of optic disc that does not pose nerve fibers. In glaucoma, the decomposition of the fiber nerves is caused that leads to increased cup size, thereby CDR. The CDR is an essential clinical indicator that helps to make the diagnosis in glaucoma.^{20,21} The vertical CDR increases tremendously in the initial stage as well as in the intermediate stage.⁹

OCT is proved to be helpful in diagnosing various types of glaucoma by using infrared light to compute the distance between the anatomical structures contained in the eye and is expedient.²² The measurement of ACA is utilized for detecting the ACG. Thus, the evaluations can be envisaged and used in AS-OCT imaging methods. AS-OCT offers outstanding reproducibility and repeatability for computing depth, anterior chamber angle, and corneal thickness. It is noted that the ACG is the result of one or more methods for detecting the anterior eye segment. Moreover, patients with different ACG methods lead to variations in the measurements of the anterior

segment.²³ The analysis of the features dimensions generated by the AS-OCT and their classification is helpful for diagnosing ACG.²⁴

The aim is to present a novel optimization-driven strategy, namely, Jaya-CSO-based RNN for Glaucoma detection. The proposed Jaya-CSO is developed by incorporating the Jaya algorithm in the CSO algorithm for training the RNN classifier. Initially, the input image is fed to the preprocessing for removing noise and artifacts present in the image. The obtained preprocessed image is fed to both optic disc detection and blood vessel detection concurrently. In optic disc detection, the binarization, and circle fitting technique is applied for extracting the significant features, and simultaneously in blood vessel detection, the Renyi entropy-based sparking process is applied for extracting the essential features for the classification. In addition, the image level features are obtained from the preprocessed image. Finally, the image level features, optic disc level features, and blood vessel features are extracted, transformed as a feature vector. The obtained feature vector is subjected to the RNN classifier, wherein the proposed Jaya-CSO is employed for training the RNN for glaucoma detection. The proposed method is effective in determining glaucoma present in the input image.

1.1 | The major contribution of the paper

1.1.1 | Proposed Jaya-CSO-based RNN for glaucoma detection

The proposed Jaya-CSO is utilized for training the RNN to acquire the optimal weights for glaucoma detection. The proposed Jaya-CSO is developed by integrating the Jaya algorithm in the CSO algorithm for training RNN to detect glaucoma contained in the input retinal image.

The paper is structured as follows Section 1 elaborates on the introductory part of the glaucoma detection techniques using the retinal fundus images. Section 2 deliberates the conventional techniques based on glaucoma detection, and section 3 elaborates the proposed glaucoma detection framework using the Jaya-CSO-based RNN model, and Section 4 demonstrates experimental results attained by the proposed method, and section 5 deliberates the conclusion.

2 | LITERATURE SURVEY

The section deliberates the literature survey of the glaucoma detection techniques along with their advantages

and disadvantages. At the end of the section, the challenges of the conventional techniques are portrayed.

Mane and Jadhav⁵ developed a framework named sparking process and a holoentropy-based decision tree for the automatic classification of DR for enhancing the effectiveness. The method employed a sparking algorithm for automatically segmenting the blood vessels by evaluating an optimal threshold. The holoentropy enabled decision tree was designed for automatically classifying the DR images into normal or abnormal based on hybrid features that preserve the disease-level patterns using the signal levels of the feature. However, the method was inapplicable with advanced features.

Phan et al²⁵ used DCNNs for differentiating glaucoma regions using color fundus images. The glaucoma detection was carried out using a set of color fundus images and the factors based on the discriminative ability. The detection revealed that the image quality influences the discriminative ability. Thus, the collection of high quality and benchmark images was essential for better detection. The method was effective in diagnosing glaucoma in its earlier stage, but the method was not precise. Moreover, the artificial intelligence system should be integrated for precise glaucoma detection.

Yu et al⁶ developed a U-Net architecture using 335 pretrained ResNet-34 as encoding down-sampling layers for segmenting optic disc and cups. The model was trained using the epochs for glaucoma detection. The method showed robust performance but required more performance parameters for precise detection. Gour and Khanna⁷ designed an automated diagnostic system based on fundus images for detecting glaucoma disease. The method focused on the mining of PHOG and GIST features using preprocessed fundus images. The mined images are ranked and chosen through PCA for selecting efficient features. The classification of glaucomatous images is carried out using the SVM classifier. However, the method failed to use deep learning models for classifying the glaucomatous and nonglaucomatous fundus images.

Kirar and Agrawal⁸ developed a hybrid approach by combining the DWT and EWT for glaucoma disease identification using fundus images. This method extracted the green channel image from the fundus images and then decomposed the image using EWT-DWT, DWT, DWT-EWT, and EWT. The method extracted 77 features and subjected the obtained features to the LS-SVM classifier with RBF kernel. The method was effectual in diagnosing glaucoma with enhanced results. However, few parameters were considered for the evaluation, which could not detect the overall performance.

Ramzan et al¹⁰ developed a method for extracting the RPE layers and ILM from the OCT images. The method

utilized color channels for evaluating the quality and was applied for segmenting the ILM layer for different quality images. In this method, the centroid based thresholding was utilized for removing the ILM regions. This method used RPE breakpoints and the ILM layer for computing the optic discs. Based on the computed CDR, the system classified the image as normal or glaucomatous. The method failed to generate automated algorithm by combining the method with visual field tests.

Khalil et al¹¹ developed a computer-aided diagnosis system using hybrid structural and textural features. The system facilitated effective decisions by evaluating different conditions of glaucoma. The method contained two modules, namely, HTF and HSF. The HSF module helped to classify an image using SVM with different conditions of structural glaucoma, and the HTF module analyzed the images using different intensity and texture-based features along with SVM for taking decisions. However, the method was unable to provide precise results due to fundus images. Thus, the combination of fundus and OCT images can be done for generating precise glaucoma detection.

Salam et al²⁶ developed a hybrid structural and non-structural features for enhancing the accuracy in detecting the glaucoma disease. The structure features utilized cup to disc ratio, and the nonstructural features utilized texture and intensity for the glaucoma diagnosis. The method provided an algorithm for detecting glaucoma by analyzing the changes of structures using a set of fundus images. The results were employed by the machine learning module for making decisions. The hybridization of machine learning and structural changes improved the accuracy and sensitivity parameter. Moreover, the method was unable to screen the glaucoma patients with high accuracy, thereby led to high computational complexities in the system.

Benzebouchi et al²⁷ developed a multimodal classification method for diagnosing glaucoma based on DCNN and SVM classifiers, which automatically generate the most informative features. This method gave a highly robust and accurate result, but it did not focus on textual information. Abiodun et al²⁸ did a comprehensive review of PR, which showed ANN was the best comparable method using in statistical, fuzzy, structural techniques to address the PR problems. Abiodun et al²⁹ did a survey based on the ANN contribution, which accuracy, latency, volume, processing speed, performance, fault tolerance, and scalability was high.

Raghavendra et al³⁰ developed the CNN model, which could efficiently detect the class of unknown images to avoid the sophisticated design of hand-crafted features. Anyhow, this method was not advisable for earlier stage detection of glaucoma. Al-Bander et al³¹

developed an automated approach based on CNN, which diagnosing glaucoma with considerably lower computational cost, but it was only applicable for the smaller dataset. Gómez-Valverde et al³² developed glaucoma detection based on the CNN approach, which was applicable for the large scale datasets, but the classification carried out only on color fundus images. Diaz-Pinto et al³³ developed an automatic glaucoma assessment model based on CNN, in which the classification accuracy was high, but a different labeling criterion was the major drawback of this method.

2.1 | Challenges

- Considering different cases of existing methods with progressive conditions, it is impossible to differentiate glaucomatous and nonglaucomatous variation of the optic disc with a single examination. Moreover, the OCT analysis of optic nerve head may result in performance overhead.³⁴
- The myopia is a risk factor of glaucoma, which can lead to functional and structural defects that cannot be corrected in glaucoma detection.³⁵
- In Ramzan et al,¹⁰ optic cup to disc ratio and retinal layers segmentation is utilized for detecting glaucoma. However, the correlated result is offered with more sensitivity, and the thickness of the OCT retinal layer was not evaluated.
- The fundus images are used for glaucoma detection by examining the optic nerves. However, the fundus images may provide inaccurate results due to the high-intensity variations.¹¹
- In Gour and Khanna,⁷ GIST and PHOG descriptors are utilized for glaucoma detection using blood vessel segmentation. However, the inaccurate segmentation may lead to erroneous calculation and increase the misclassification rates.
- The detection of the optic disc is a major challenge in disease diagnostic systems due to the variations of structures and interruption of blood vessels. In addition, the low-resolution image makes the segmentation process complex with an imprecise diagnosis.

3 | PROPOSED JAYA-CSO BASED RNN FOR GLAUCOMA DETECTION

Glaucoma is a visual condition whose succession results in perpetual blindness. Glaucoma is a chronic disease wherein the succession is stopped only if timely detected. The conventional methods are time-consuming and need more labor, cost, and time. To conquer the issues facade

by the conventional techniques, the automatic detection of glaucoma patients from their retinal fundus images using the Jaya-CSO based RNN is proposed wherein optic disc, an input image, and blood vessel features are extracted to perform the classification. Initially, the input retinal fundus images are preprocessed for making the image appropriate for consequent processing.

The preprocessed images are subjected to optic disc detection module and blood vessel detection module simultaneously. In optic disc segmentation, the input retinal fundus image undergoes segmentation using circle fitting and binarization. Subsequently, the blood vessel segmentation is done from the preprocessed image through the sparking process. In the sparking process, Renyi entropy³⁶ is used for effective information gain. Then, the statistical features, and the image level features are extracted from the optic disc and the blood vessels, whereas the features, such as B-spline coefficients,³⁷ LOOP,³⁸ LGP,³⁸ and FFT coefficients³⁷ is extracted directly via the preprocessed image. The extracted features obtained from the optic disc, blood vessels, and the fundus image is formulated in a feature vector.

Finally, the glaucoma classification is performed with RNN³⁹ using the feature vector wherein the RNN is trained using the proposed Jaya-CSO. The proposed Jaya-CSO is the integration of the standard Jaya algorithm⁴⁰ and CSO⁴¹ to inherit the merits of both the optimizations, toward the effective training of the classifier. Figure 1 portrays the glaucoma detection framework using proposed Jaya-CSO based RNN.

Assume a database D with p number of retinal fundus images and is formulated as,

$$D = \{I_1, I_2, \dots, I_k, \dots, I_p\} \quad (1)$$

where I_k represents the k th input image, and p represent total images.

3.1 | Preprocessing using input fundus images

The importance of preprocessing is to enable smoother processing of the input image. Preprocessing is considered an essential step for processing the images to make it suitable for the detection process. Additionally, the preprocessing is carried out for eliminating the noise contained in the image. Moreover, the preprocessing can be aided as an image enhancement module, which poses the capability to enhance the contrast of the image for effective glaucoma detection.

The preprocessed images are fed to the optic disc detection module as well as the blood vessel detection

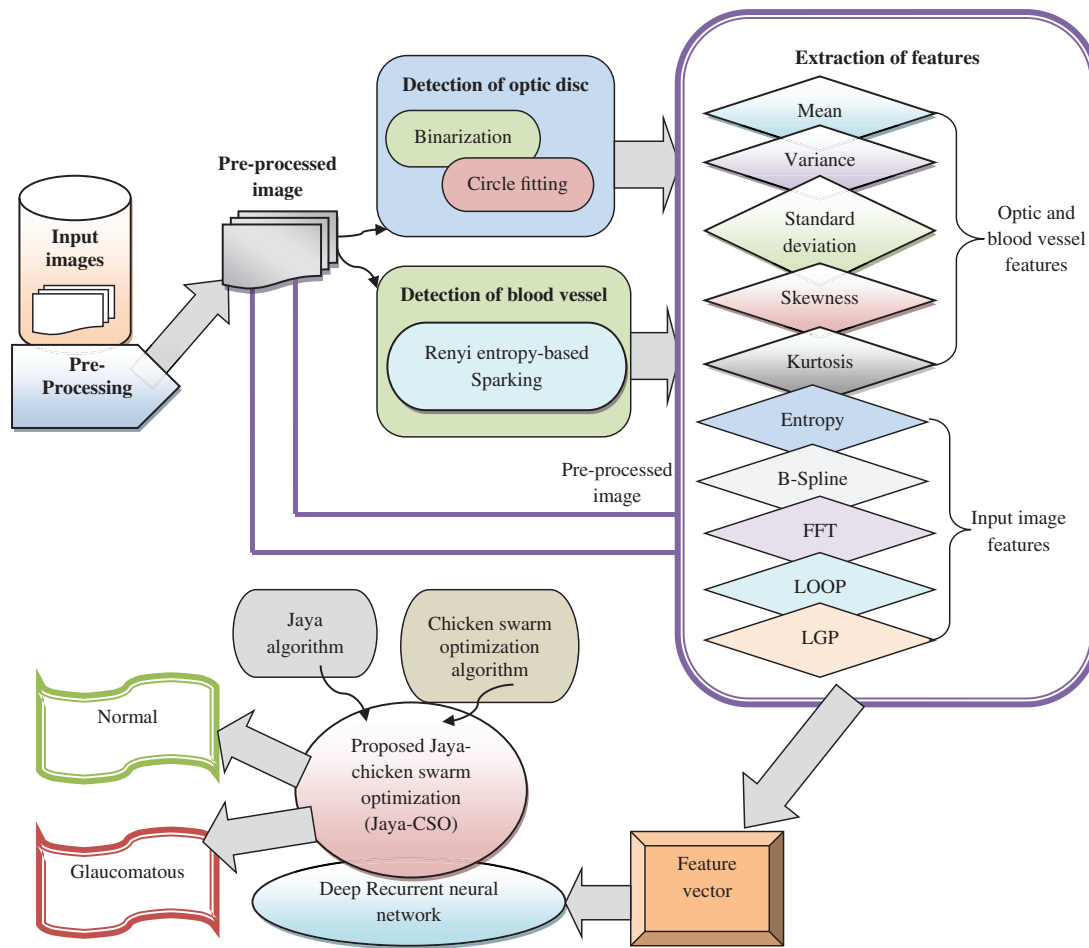


FIGURE 1 Block diagram of propose glaucoma detection method using Jaya-CSO based RNN [Color figure can be viewed at wileyonlinelibrary.com]

module simultaneously for extracting the significant features that are suitable for the glaucoma detection.

3.1.1 | Detection of the optic disc using binarization and circle fitting techniques for feature extraction

Once the preprocessing is done, the detection of the optic disc is performed using binarization and circle fitting techniques. The optic disc regions pose high intensities, hard exudates, and sharp boundaries than the other image regions. Thus the determinations of the optic disc are of great use for finding the glaucomatous regions. Thus, the binarization and circle fitting techniques are applied for the optic disc detection and are illustrated below:

Binarization

The optic disc is detected using the preprocessed image by the smooth textures using the binarization techniques.

The binarization helps to separate the complicated regions from the smoother regions by evaluating the values of the pixels contained in the image. The binarization methods are utilized for converting the greyscale image ranging from 0 to 256 Gy levels into a white and black image, which can be either 0 or 1. The high-quality binary image provides a more accurate optic disc as compared to the input image because of high intensities.

In binarization, two classes result is generated using the set of images based on the value of each pixel, wherein the pixel is assigned either a value 1 or the value 0 based on the value of the mean of an entire pixel. If the mean value is greater, then the value is 1 or else the value is 0. Moreover, the binarization method uses thresholding for generating binary images. In thresholding, if the pixel value is larger than the threshold then the pixel is an object with 1 as value else the pixel is a background pixel with value as 0. Finally, based on the label of each pixel, a binary image is constructed by filling each pixel with white or black color.

Circle fitting

The circle fitting to a given set of points in the plane is a major area like statistics and medical image processing. Here, the circle fitting is done by fitting the circle around the region of the optic disc. The optic disc shape is utilized for classifying the images for detecting the normal and glaucomatous retinal images.

The attributes considered in the circle fitting for the optic disc are (a) R is the radius of the circle, (b) Y is the y -coordinate of circle, and (c) X is the x -coordinate of circle.

3.1.2 | Detection of blood vessel regions using Renyi entropy-based sparking process for extracting features

In blood vessel segmentation, the sparking algorithm is adapted for segmenting the blood vessels by estimating optimal threshold values.

Here, the preprocessed image I'_k of size $x \times p$ endures a sparking process. An exponential kernel is adapted for converting the input image into p different images. Consider h be a random number produced with a size of $1 \times p$ and K_d be the threshold value given by the user.

The exponential kernel is determined by transforming the data into the exponential form as given in Equation (2).

$$l_1 = \exp \left[\frac{h^2}{2K_d^2} \right] \quad (2)$$

The vector l_1 is normalized for obtaining the kernel, which is formulated as,

$$f_1 = \frac{\left[\frac{1}{p} \sum_{k=1}^p \max(l_1) - l_1 \right] - l_1}{\sum_{k=1}^p \max(l_1) - l_1} \quad (3)$$

The input preprocessed image is rotated in p different angles to generate p different images of similar size. Finally, the sparking image is generated, which is given as,

$$\alpha_{\text{new}}(k, m) = \frac{\alpha_l + \alpha_{\min}(k, m)}{\alpha_{\max}(k, m)} 255 \quad (4)$$

where $\alpha_{\min}(k, m) = \min[s_1(k, m), s_2(k, m), \dots, s_p(k, m)]$, and $\alpha_{\max}(k, m) = \max[s_1(k, m), s_2(k, m), \dots, s_p(k, m)]$.

The search of the optimum threshold is performed using the sparking image. The neighbor pixel matrix is generated in which each row defines the pixel values of the current pixel, and each column states the pixel values of the neighbor pixel. Each pixel from value 0 to 255 is evaluated using a function called *fun* for computing the optimum threshold. Consider a threshold B , then the E and F matrix are formed as,

$$E \Leftarrow [B \times B] \quad (5)$$

$$F \Leftarrow [(255 - B) \times (255 - B)] \quad (6)$$

The entropy of E and F is computed for the information gain for threshold B , which is done with the Renyi entropy.³⁶ The Renyi entropy of E_k is formulated as,

$$R_e(E) = \frac{1}{1-\alpha} \log \left(\sum_{k=1}^p P(E_k)^j \right) \quad (7)$$

where j indicates the real number, p specifies the total images; $1 \leq k \leq p$, $P(E_k)$ is the probability of E_k .

The Renyi entropy of F_k is formulated as,

$$R_e(F) = \frac{1}{1-\alpha} \log \left(\sum_{k=1}^p P(F_k)^j \right) \quad (8)$$

where j indicates the real number, p specifies the total images; $1 \leq k \leq p$, $P(F_k)$ is the probability of F_k , and the Renyi entropies of E and F are symbolized as $R_e(E)$ and $R_e(F)$.

The summation of the entropies provides the selection of optimal threshold value.

$$G = R_e(E) + R_e(F) \quad (9)$$

3.1.3 | Extraction of features for generating a feature vector using optic disc features and blood vessel features

The extraction of features guarantees effective glaucoma detection for which the optic disc, blood vessel, and image-level features are adapted. For the features extraction, the optic disc, blood vessel, and the preprocessed image are adapted in such a way that the features assure improved accuracy in classification. The features extracted from the image are illustrated below:

(a) *Mean*: The mean is calculated by computing the average of pixels contained in the image that are formulated as,

$$\mu = \frac{1}{|q(S_n)|} \times \sum_{n=1}^{|q(S_n)|} q(S_n) \quad (10)$$

where n is the total segments, $q(S_n)$ indicate the values of the pixel of each segment and $|q(S_n)|$ indicates the total pixels contained in the segment.

(b) *Variance*: The variance feature σ is calculated based on the mean value that is expressed as

$$\sigma = \frac{\sum_{n=1}^{|q(S_n)|} |S_n - \mu|}{q(S_n)} \quad (11)$$

(c) *SD*: The SD is the square root of the variance, which is denoted by ρ .

(d) *Kurtosis and skewness*: Kurtosis κ represents the evenness that defines the sharpness of the peak. The connotation of kurtosis and skewness δ is that it describes object shape based on the arithmetical value. The skewness represents the relative symmetry, whereas kurtosis represents the relative peakedness of the probability distribution.

(e) *Entropy*: The entropy⁴² is a standard measure used for determining the uncertainty in any data and is utilized for maximizing mutual information in different operations. The wide availability of entropy variations inspired the suitability preference for a particular operation. Thus, the entropy of an image is used for targeting the difference between neighbor pixels or a pixel group. Moreover, the entropy is described as corresponding states of intensity level that the individual pixels can adapt. The entropy of an image pixel can be used in quantitative analysis and for evaluating the details of an image and provides a better comparison between the image details.

Thus, the entropy is computed for the obtained probability, which is expressed as,

$$\varepsilon = -P \log(P) \quad (12)$$

where P denotes the probability distribution of pixels in an image.

(f) *LGP*: The LGP³⁸ is utilized for obtaining the texture features using the value of gradient using eight neighboring pixels based on the absolute difference of intensities between the center pixels and the neighboring pixel. In this feature, the threshold is computed by taking gradient values average and $2 \times h + 1$ kernel are adapted for calculating the neighboring pixels of the center pixel. If the pixel's gradient is higher than the threshold, then the binary value of the neighboring

pixel is termed as 1 or else it is 0. The LGP operator is given as,

$$\vartheta = \text{LGP}_{\ell, v}^{x^u, p^v} = \sum_{i=0}^{l-1} \lambda[\varpi_{\ell} - \varpi^*] \times 2^i \quad (13)$$

$$\lambda(o) = \begin{cases} 0; & o < 0 \\ 1; & \text{otherwise} \end{cases} \quad (14)$$

where ϖ^* represent the average of the gradient values and ϖ_{ℓ} indicates the values of gradient between the center pixel (w^*) located at (x^u, y^u) and its neighboring pixels (w) and it is given as,

$$\varpi_l = |w - w^*|. \quad (15)$$

The above equation defined that the LGP produces the codes by thresholding the absolute intensity difference based on the locally adapted threshold ϖ^* . The LGP produces the invariant patterns based on gradient difference, and LGP does not affect due to the local color variations.

(g) *LOOP*: The LOOP³⁸ feature is the integration of LBP and LDP features that help to overcome the drawbacks of LDP and LBP. The major issue in LBP and LDP represents the arbitrary sequence of binarization weights, which adds a dependency on the orientation. The LOOP tends to be a novel feature descriptor that could capture the local intensity variations of different image patterns. The LOOP feature is formulated as,

$$M = \text{loop}(x^u, p^u) = \sum_{i=0}^7 z(c_i - c_u) \cdot 2^{w_i} \quad (16)$$

$$z(x) = \begin{cases} 1; & x \geq 0 \\ 0; & \text{otherwise} \end{cases} \quad (17)$$

(h) *Fast Fourier transforms*: Based on the intensities of the spatial pixel, the Fourier coefficients compute the frequency information of an image and are computed using the Fourier Transform. The FFT is applied to the preprocessed image I_k and compress the responses using the principal component analysis, and the result after applying FFT is given by H .

(i) *B-Spline coefficients*: The B-spline coefficients decode the spatial frequencies of the information based on the piecewise polynomials through a pixel neighborhood.

The input image of I_k of size $x \times p$ with $u' = (1...e)$ and $v' = (1...r)$ is converted to B-spline coefficients $C'(s, s')$ with $s = \{1, ...e\}$ and $s' = \{1...r\}$.

$$N = p(u', v') = \sum_{s=0, s'=0}^{x, p} C(s, s') \alpha^m(u-s)(v-s') \quad (18)$$

where m is the degree of central B-spline, $\alpha^m(u, v)$ indicates the central of B-spline.

The B-spline coefficients are computed from the preprocessed image I_k and is given by N .

3.1.4 | Formation of the feature vector

Equation (12) demonstrates the blood vessel, optic disc, and input image feature. Thus, the features extracted are given as,

$$J = \{\mu, \sigma, \rho, \kappa, \delta, \varepsilon, \vartheta, H, M, N\} \quad (19)$$

where J indicates the feature vector that is extracted using the input image, blood vessel, and optic disc, $\mu, \sigma, \rho, \kappa, \delta$ and ε indicates mean, variance, SD, kurtosis, skewness, and entropy are the blood vessel and optic disc features, whereas the image level features are LGP, LOOP, FFT, and B-Spline that are denoted as ϑ, H, M , and N . The feature vector is provided to the RNN, which classifies the input images based on features provided and derives the class label. The classifier derives the class label and classifies the normal and glaucomatous corresponding to the input image.

3.2 | Glaucoma detection using the proposed Jaya-CSO-based RNN

In this section, the Glaucoma detection using the proposed Jaya-CSO method is presented, and the detection is progressed using the feature vector. The extracted features are presented for the classification using RNN,³⁹ and the training of the classifier is done through the proposed training algorithm, called Jaya-CSO, which is the integration of the Jaya algorithm⁴⁰ with the CSO.⁴¹ The goal of the proposed Jaya-CSO is to identify the input image based on the extracted features from the blood vessels and the optic disc for evaluating whether the image is normal or glaucomatous. The section deliberates the architecture of RNN, and the algorithmic steps of the proposed Jaya-CSO are described below.

3.2.1 | Architecture of RNN

RNN³⁹ is a type of supervised machine learning model that is composed of artificial neurons with one recurrent loop known as the feedback loop. The feedback loops refer to the recurrent cycles that run overtime in a

sequential manner. The training of RNN is done in a supervised fashion, which requires a training dataset that consists of input-target pairs. The goal of the RNN is to reduce the difference between the output and the target groups by adapting the weight optimization procedure in the network. RNN is capable of processing the sequential inputs by adapting a recurrent hidden state, whose activation is based on the previous step.

In this way, the network exhibits dynamic temporal behavior. Moreover, the recurrent layer permits feeding the information of previous time steps and combines the resultant output with the input of the current time steps, which means the order of input information plays a vital role in processing. The algorithm permits continuous improvement of required output and thus, makes the network to be more intelligent with each new update. In addition, RNN is applicable in several fields in which the data is represented in sequence.

RNN is composed of three layers, which are the input layer, recurrent hidden layer, and output layer, as depicted in Figure 2.

The input layer is composed of a sequence of vectors with time u , such as $\{..., J_{u-1}, J_u, J_{u+1}, ...\}$, where J_u , which represents the selected features obtained based on the proposed Jaya-CSO algorithm. The input of RNN is connected with the hidden units in the hidden layer in which the connections are described using the weight matrix $\omega_{I, H}$. The hidden layer has W hidden units such that $C_u = (C_1, C_2, ...C_w)$. The hidden layer describes the state space or system memory as,

$$C_u = \alpha(\gamma_u). \quad (20)$$

From the above equation, the formula of state space of system memory γ_u is given as,

$$\gamma_u = \omega_{IH} J_u + \omega_{HH} J_{u-1} + \beta_h \quad (21)$$

where γ_u is the system memory, α refers the activation function, β_h denote the bias vector of hidden units, ω_{IH} is the weight connecting the input and hidden layers, ω_{HH} refers to the weight linking the two hidden layers. Thus, the output vector is given by,

$$O_u = \alpha'(\omega_{HO} h_{u-1} + \beta_o) \quad (22)$$

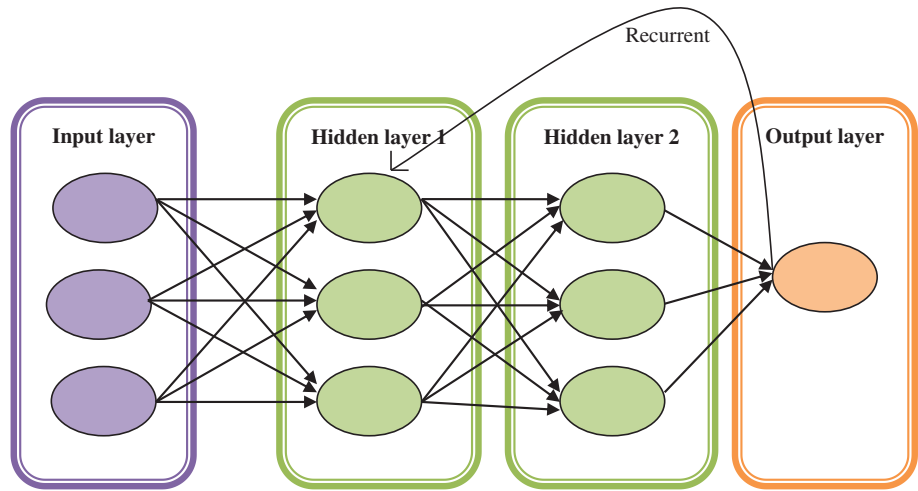
where β_o is the bias of output layer and α is the activation function.

3.2.2 | Training of RNN

The training of RNN³⁹ is carried out using the proposed Jaya-CSO optimization algorithm that intends to

FIGURE 2 Architecture of RNN

[Color figure can be viewed at
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determine the optimal weights for tuning the RNN classifier for glaucoma detection. The brief illustration of the algorithmic steps carried out in the execution of the proposed Jaya-CSO algorithm is described in this section. The optimal weights are devised from the proposed Jaya-CSO algorithm, which helps to tune the RNN for deriving the optimal classification results. The glaucoma detection adapts the proposed Jaya-CSO-based RNN for classifying the input image by devising an optimal classification and is able to deal with the new images that are arrived from the distributed sources. The algorithmic steps of the proposed Jaya-CSO are illustrated below:

Step 1: Initialization

The first step is the initialization of the chicken population, along with its related parameters.

$$A = \{A_1, A_2, \dots, A_a, \dots, A_v\} \quad (23)$$

where A_a is the position of a th chicken, and v is the total number of population.

Step 2: Evaluation of error

The best solution is determined based on the fitness function, which is termed as the minimization issue, and thus, the solution generating less mean square error (MSE) is chosen as the best solution. Here, the MSE is calculated as follows,

$$MS_{err} = \frac{1}{c} \sum_{o=1}^c [M_o - M_o^*]^2 \quad (24)$$

where M_o is the expected output and M_o^* is the predicted output, c specifies the number of data samples, where $1 < o \leq c$.

Step 3: Determination of update equation

For maximizing the speed of convergence rate, the CSO algorithm is being utilized. The CSO algorithm is highly proficient and provides optimal performance while computing the solutions, and there is less parameter needed for fine-tuning, and the CSO algorithm is used due to its simpler algorithmic structure. As per the CSO algorithm,⁴¹ the solution update is expressed as,

$$A_{a,t}(o+1) = A_{a,t}(o) * (1 + \text{Randn}(0, \eta^2)) \quad (25)$$

where η is the SD, $A_{a,t}(o+1)$ is the position of a th chicken in $(o+1)$ th iteration in t th dimension, $A_{a,t}(o)$ is the position of a th chicken in (o) th iteration in t th dimension, $\text{Randn}(0, \eta^2)$ is the Gaussian distribution with mean as 0 and SD as η^2 .

To enhance the algorithmic performance and to solve the constrained and unconstrained optimization issues, the Jaya algorithm is utilized. Thus, the update equation based on the Jaya algorithm⁴⁰ is given as,

$$A_{a,t}(o+1) = A_{a,t}(o) + Q_{1,a}(o)(A_{a,best}(o) - |A_{a,t}(o)|) - Q_{2,a}(o)(A_{a,worst}(o) - |A_{a,t}(o)|) \quad (26)$$

where $A_{a,t}(o)$ is the current solution, $A_{a,t}(o+1)$ is the next solution, $Q_{1,a}(o)$ and $Q_{2,a}(o)$ is the random number, and $A_{a,best}$ and $A_{a,worst}(o)$ represent the best and worst solution.

If $A_{a,t}(o)$ is positive

$$A_{a,t}(o+1) = A_{a,t}(o) + Q_{1,a}(o)(A_{a,best}(o) - A_{a,t}(o)) - Q_{2,a}(o)(A_{a,worst}(o) - A_{a,t}(o)) \quad (27)$$

$$A_{a,t}(o+1) = A_{a,t}(o) [1 - Q_{1,a}(o) + Q_{2,a}(o)] + Q_{1,a}(o)A_{a,best}(o) - Q_{2,a}(o)A_{a,worst}(o) \quad (28)$$

$$A_{a,t}(o) [1 - Q_{1,a}(o) + Q_{2,a}(o)] = A_{a,t}(o+1) - Q_{1,a}(o)A_{a,best}(o) + Q_{2,a}(o)A_{a,worst}(o) \quad (29)$$

$$A_{a,t}(o) = \frac{A_{a,t}(o+1) - Q_{1,a}(o)A_{a,best}(o) + Q_{2,a}(o)A_{a,worst}(o)}{[1 - Q_{1,a}(o) + Q_{2,a}(o)]} \quad (30)$$

Thus, the solution update of the proposed Jaya-CSO algorithm after substituting Equation (30) in Equation (25) is expressed as,

$$A_{a,t}(o+1) = \left[\frac{A_{a,t}(o+1) - Q_{1,a}(o)A_{a,best}(o) + Q_{2,a}(o)A_{a,worst}(o)}{[1 - Q_{1,a}(o) + Q_{2,a}(o)]} \right] * (1 + \text{Randn}(0, \eta^2)) \quad (31)$$

$$A_{a,t}(o+1) = \frac{A_{a,t}(o+1)}{1 - Q_{1,a}(o) + Q_{2,a}(o)} + \frac{A_{a,t}(o+1)}{1 - Q_{1,a}(o) + Q_{2,a}(o)} \text{Randn}(0, \eta^2) - \frac{Q_{1,a}(o)A_{a,best}(o) + Q_{2,a}(o)A_{a,worst}(o)}{1 - Q_{1,a}(o) + Q_{2,a}(o)} (1 + \text{Randn}(0, \eta^2)) \quad (32)$$

$$A_{a,t}(o+1) - \frac{A_{a,t}(o+1)}{1 - Q_{1,a}(o) + Q_{2,a}(o)} - \frac{A_{a,t}(o+1)}{1 - Q_{1,a}(o) + Q_{2,a}(o)} \text{Randn}(0, \eta^2) = - \frac{Q_{1,a}(o)A_{a,best}(o) + Q_{2,a}(o)A_{a,worst}(o)}{1 - Q_{1,a}(o) + Q_{2,a}(o)} (1 + \text{Randn}(0, \eta^2)) \quad (33)$$

$$A_{a,t}(o+1) \left[1 - \frac{1}{1 - Q_{1,a}(o) + Q_{2,a}(o)} - \frac{\text{Randn}(0, \eta^2)}{1 - Q_{1,a}(o) + Q_{2,a}(o)} \right] = - \frac{Q_{1,a}(o)A_{a,best}(o) + Q_{2,a}(o)A_{a,worst}(o)}{1 - Q_{1,a}(o) + Q_{2,a}(o)} (1 + \text{Randn}(0, \eta^2)) \quad (34)$$

$$A_{a,t}(o+1) \left[\frac{1 - Q_{1,a}(o) + Q_{2,a}(o) - 1 - \text{Randn}(0, \eta^2)}{1 - Q_{1,a}(o) + Q_{2,a}(o)} \right] = - \frac{Q_{1,a}(o)A_{a,best}(o) + Q_{2,a}(o)A_{a,worst}(o)}{1 - Q_{1,a}(o) + Q_{2,a}(o)} (1 + \text{Randn}(0, \eta^2)) \quad (35)$$

$$A_{a,t}(o+1)Q_{1,a}(o) - Q_{2,a}(o) + \text{Randn}(0, \eta^2) = Q_{1,a}(o)A_{a,best}(o) - Q_{2,a}(o)A_{a,worst}(o)(1 + \text{Randn}(0, \eta^2)) \quad (36)$$

The update is performed using the algorithms, CSO and Jaya, and is evaluated in such a way that the weights with less error are employed for training RNN as per the equation (37) below,

$$A_{a,t}(o+1) = \frac{Q_{1,a}(o)A_{a,best}(o) - Q_{2,a}(o)A_{a,worst}(o)(1 + \text{Randn}(0, \eta^2))}{Q_{1,a}(o) - Q_{2,a}(o) + \text{Randn}(0, \eta^2)} \quad (37)$$

Step 4: Re-computation of weights based on error

The error is recomputed based on weights given in Equation (24). The algorithm generating minimum weight is utilized for training the RNN for glaucoma detection.

Step 5: Determination of optimum weight using the proposed Jaya-CSO-training algorithm

The weights re-updated using the proposed Jaya-CSO algorithm and are evaluated in such a way that the weights with less error are employed for training RNN as per the equation (37).

Step 6: Terminate

The optimal weights are derived iteratively until the maximum number of iterations.

4 | RESULTS AND DISCUSSION

The analysis of the proposed Jaya-CSO with other existing techniques is instantiated in this section. The efficiency is examined by evaluating the techniques based on sensitivity, accuracy, and specificity measures with the size of images as 256 and 512.

4.1 | Experimental setup

The method is executed on Windows 10 OS with 2GB RAM and Intel Core processor and is implemented in MATLAB.

4.1.1 | Experimental results

Figure 3 illustrates the experimental result of proposed Jaya-CSO for glaucoma detection. Figure 3A) portrays the set of input retinal images from the retinal fundus image database and Figure 3B) portrays the LGP extracted image using the proposed Jaya-CSO algorithm and Figure 3C) displays the segmented image of the provided input image.

4.2 | Dataset description

HRF Image dataset⁴³ is donated by the collaborative research group for supporting automatic segmentation techniques using retinal fundus images. The dataset consists of 15 healthy patient images, along with 15 patient's diabetic retinopathy images and 15 glaucomatous patients' images.

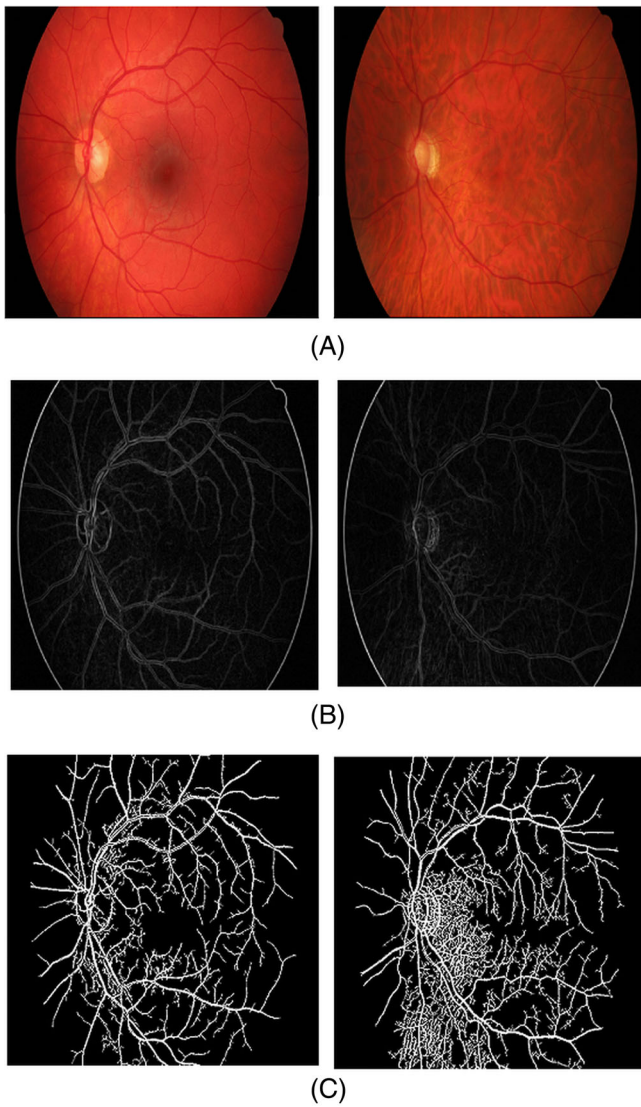


FIGURE 3 Experimental result of proposed Jaya-CSO for glaucoma detection using (A) Input retinal image, (B) LBP applied image, and (C) Segmented image [Color figure can be viewed at wileyonlinelibrary.com]

4.3 | Competing methods

The methods, such as DCNN,²⁵ RNN,⁴⁴ Holoentropy based decision tree,⁵ DeepRNN + Jaya (Applied Jaya algorithm⁴⁰ in RNN⁴⁴) and are compared with the proposed Jaya-CSO + RNN to enhance the dominance of the proposed technique.

4.4 | Performance metrics

The consideration of the proposed glaucoma detection method is done with three measures, namely accuracy, specificity, and sensitivity.

4.4.1 | Accuracy

The accuracy is described as the degree of closeness of estimated value with respect to its original value in the glaucoma detection process, and it is represented as,

$$\text{Accuracy} = \frac{T^p + T^n}{T^p + T^n + F^p + F^n} \quad (38)$$

where T^p represents the true positive T^n represents the true negative, F^p is the false positive, and F^n specifies the false negative, respectively.

4.4.2 | Sensitivity

It refers to the fraction of positives which are recognized by the glaucoma detection technique correctly, and it is formulated as,

$$\text{Sensitivity} = \frac{T^p}{T^p + F^n} \quad (39)$$

4.4.3 | Specificity

It refers to the ratio of negatives identified using the glaucoma detection technique correctly and is formulated as follows.

$$\text{Specificity} = \frac{T^n}{T^n + F^p} \quad (40)$$

4.5 | Performance analysis

The analysis of the performance of the proposed Jaya-CSO + RNN with sensitivity, accuracy, and specificity parameter is described in this section. The analysis is done with image size 256 and 512. The analysis is performed by altering the percentage of training data from 50% to 90%.

4.5.1 | Analysis with image size = 512

Figure 4 deliberates the performance analysis of proposed Jaya-CSO + RNN with varying hidden layers ranging from 5 to 25. The analysis is performed by altering the percentage of training data with image size 512. Figure 4A) shows the analysis of proposed Jaya-CSO + RNN using the accuracy parameter. For 50% training data, the accuracy computed by proposed Jaya-CSO + RNN with hidden layers = 5 is

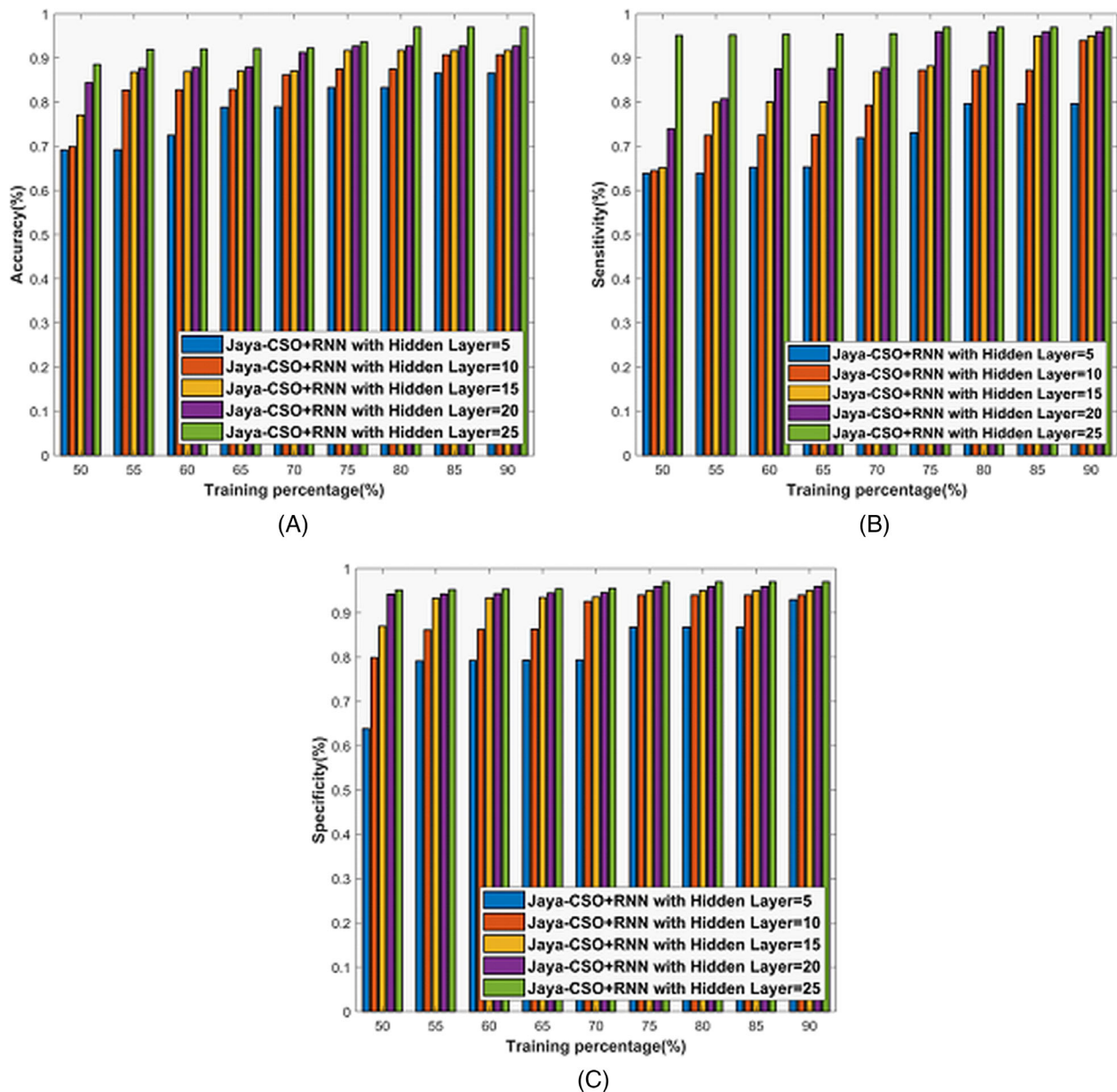


FIGURE 4 Analysis of proposed Jaya-CSO with image size 512 using (A) accuracy, (B) sensitivity, and (C) specificity [Color figure can be viewed at wileyonlinelibrary.com]

0.692, hidden layer = 10 is 0.699, hidden layer = 15 is 0.771, hidden layer = 20 is 0.844, and hidden layer = 25 is 0.885, respectively. Likewise, for 90% training data, the accuracy computed by proposed Jaya-CSO + RNN with hidden layers = 5 is 0.865, hidden layer = 10 is 0.907, hidden layer = 15 is 0.917, hidden layer = 20 is 0.926, and hidden layer = 25 is 0.97, respectively.

Figure 4B portrays the analysis of proposed Jaya-CSO + RNN based on sensitivity parameter. For 50% training data, the sensitivity values computed by proposed Jaya-CSO + RNN with hidden layers = 5 is 0.638, hidden layer = 10 is 0.645, hidden layer = 15 is 0.652, hidden layer = 20 is 0.739, and hidden layer = 25 is

0.951, respectively. Likewise, for 90% training data, the sensitivity values computed by proposed Jaya-CSO + RNN with hidden layers = 5 is 0.797, hidden layer = 10 is 0.94, hidden layer = 15 is 0.95, hidden layer = 20 is 0.96, and hidden layer = 25 is 0.97, respectively. Figure 4C shows the analysis of proposed Jaya-CSO + RNN based on specificity parameter. For 50% training data, then the specificity values computed by proposed Jaya-CSO + RNN with hidden layers = 5 is 0.638, hidden layer = 10 is 0.799, hidden layer = 15 is 0.869, hidden layer = 20 is 0.941, and hidden layer = 25 is 0.951, respectively. Likewise, for 90% training data, the specificity values computed by proposed Jaya-CSO + RNN with

hidden layers = 5 is 0.93, hidden layer = 10 is 0.94, hidden layer = 15 is 0.95, hidden layer = 20 is 0.96, and hidden layer = 25 is 0.97, respectively.

4.5.2 | Analysis based on image size = 256

Figure 5 deliberates the performance analysis of proposed Jaya-CSO + RNN with varying hidden layers ranging from 5 to 25.

Figure 5A shows the analysis of proposed Jaya-CSO + RNN using accuracy parameter. For 50% training data, then the accuracy computed by proposed Jaya-CSO + RNN with hidden layers = 5 is 0.692, hidden layer = 10 is 0.794, hidden layer = 15 is 0.835, hidden layer = 20 is 0.876, and hidden layer = 25 is 0.918, respectively. Likewise, for 90% training data, the accuracy computed by proposed Jaya-CSO + RNN with hidden layers = 5 is 0.897, hidden layer = 10 is 0.907, hidden layer = 15 is 0.917, hidden layer = 20 is 0.96, and hidden layer = 25 is 0.97, respectively.

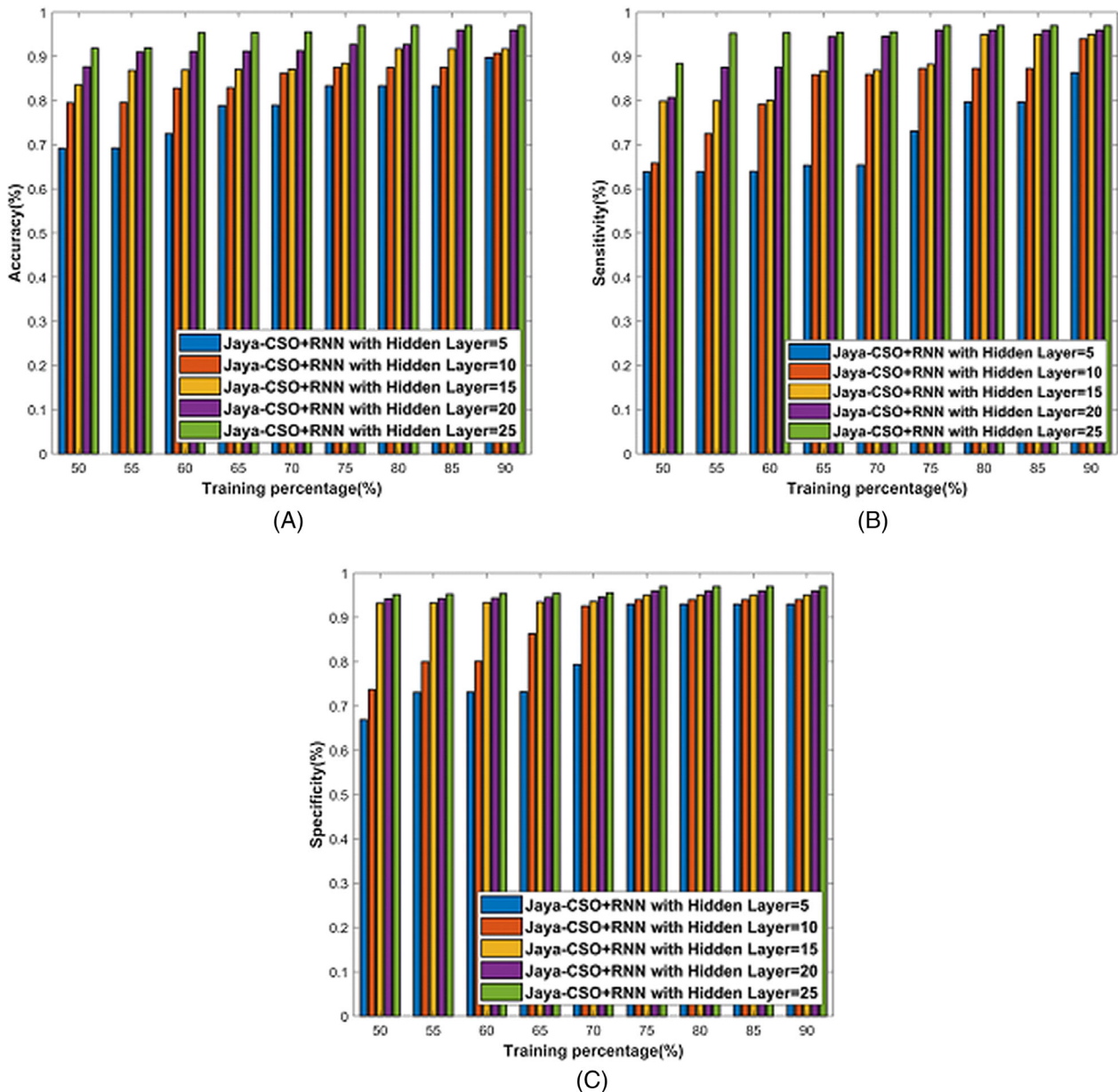


FIGURE 5 Analysis of proposed Jaya-CSO with image size 256 using (A) accuracy, (B) sensitivity, and (C) specificity [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

Figure 5B shows the analysis of proposed Jaya-CSO + RNN based on sensitivity parameter. When the training data percentage is 50, then the sensitivity values computed by proposed Jaya-CSO + RNN with hidden layers = 5 is 0.638, hidden layer = 10 is 0.658, hidden layer = 15 is 0.798, hidden layer = 20 is 0.807, and hidden layer = 25 is 0.883, respectively. Likewise, for 90% training data, the sensitivity values computed by proposed Jaya-CSO + RNN with hidden layers = 5 is 0.863, hidden layer = 10 is 0.94, hidden layer = 15 is 0.95, hidden layer = 20 is 0.96, and hidden layer = 25 is 0.97, respectively. Figure 5C shows the analysis of proposed Jaya-CSO + RNN based on specificity parameter. For 50% training data, then the specificity values computed by proposed Jaya-CSO + RNN with hidden layers = 5 is 0.669, hidden layer = 10 is 0.737, hidden layer = 15 is 0.931, hidden layer = 20 is 0.941, and hidden layer = 25 is 0.951, respectively. Likewise, for 90% training data, the specificity values computed by proposed Jaya-CSO + RNN with hidden layers = 5 is 0.93, hidden layer = 10 is 0.94, hidden layer = 15 is 0.95, hidden layer = 20 is 0.96, and hidden layer = 25 is 0.97, respectively.

4.6 | Comparative analysis

The analysis of methods with accuracy, sensitivity, and specificity parameter is described in this section. The analysis is done with image size 256 and 512.

4.6.1 | Analysis with image size = 512

Figure 6 deliberates the comparative analysis of proposed Jaya-CSO + RNN with existing methods by varying hidden layers ranging from 5 to 25.

Figure 6A shows the analysis of methods using the accuracy parameter. For 50% training data, the accuracy computed by DCNN, DeepRNN, Holoentropy-based decision tree, DeepRNN + Jaya, and proposed Jaya-CSO + RNN are 0.638, 0.731, 0.745, 0.844, and 0.885, respectively. Likewise, for 90% training data, the accuracy computed by DCNN, DeepRNN, Holoentropy-based decision tree, DeepRNN + Jaya, and proposed Jaya-CSO + RNN are 0.837, 0.907, 0.917, 0.928, and 0.97, respectively.

Figure 6B shows the analysis of proposed Jaya-CSO + RNN using the sensitivity parameter. For 50% training data, the sensitivity values computed by DCNN, DeepRNN, Holoentropy-based decision tree, DeepRNN + Jaya, and proposed Jaya-CSO + RNN are 0.638, 0.645, 0.732, 0.816, and 0.951, respectively. Likewise, for 90% training data, the sensitivity values computed by DCNN, DeepRNN, Holoentropy-based decision tree, DeepRNN +

Jaya, and proposed Jaya-CSO + RNN are 0.806, 0.94, 0.95, 0.96, and 0.97, respectively. Figure 6C shows the analysis of proposed Jaya-CSO + RNN using the specificity parameter. For 50% training data, then the specificity values computed by DCNN, DeepRNN, Holoentropy-based decision tree, DeepRNN + Jaya, and proposed Jaya-CSO + RNN are 0.638, 0.658, 0.798, 0.878, and 0.892, respectively. Likewise, for 90% training data, the specificity values computed by DCNN, DeepRNN, Holoentropy-based decision tree, DeepRNN + Jaya, and proposed Jaya-CSO + RNN are 0.863, 0.877, 0.890, 0.96, and 0.97, respectively.

4.6.2 | Analysis with image size = 256

Figure 7 deliberates the comparative analysis of proposed Jaya-CSO + RNN with existing methods by varying hidden layers ranging from 5 to 25.

Figure 7A portrays the analysis of methods using the accuracy parameter. For 50% training data, then the accuracy computed by DCNN, DeepRNN, Holoentropy-based decision tree, DeepRNN + Jaya, and proposed Jaya-CSO + RNN are 0.638, 0.731, 0.771, 0.784, and 0.885, respectively. Likewise, for 90% training data, the accuracy computed by DCNN, DeepRNN, Holoentropy-based decision tree, DeepRNN + Jaya, and proposed Jaya-CSO + RNN are 0.801, 0.814, 0.884, 0.926, and 0.937, respectively.

Figure 7B shows the analysis of proposed Jaya-CSO + RNN using the sensitivity parameter. For 50% training data, then the sensitivity values computed by DCNN, DeepRNN, Holoentropy-based decision tree, DeepRNN + Jaya, and proposed Jaya-CSO + RNN are 0.638, 0.645, 0.652, 0.816, and 0.951, respectively. Likewise, for 90% training data, the sensitivity values computed by DCNN, DeepRNN, Holoentropy-based decision tree, DeepRNN + Jaya, and proposed Jaya-CSO + RNN are 0.806, 0.877, 0.95, 0.96, and 0.97, respectively. Figure 7C shows the analysis of proposed Jaya-CSO + RNN using the specificity parameter. For 50% training data, then the specificity values computed by DCNN, DeepRNN, Holoentropy-based decision tree, DeepRNN + Jaya, and proposed Jaya-CSO + RNN are 0.638, 0.645, 0.732, 0.816, and 0.892, respectively. Likewise, for 90% training data, the specificity values computed by DCNN, DeepRNN, Holoentropy-based decision tree, DeepRNN + Jaya, and proposed Jaya-CSO + RNN are 0.730, 0.814, 0.882, 0.96, and 0.97, respectively.

4.7 | Comparative discussion

Table 1 portrays the comparative analysis of proposed Jaya-CSO + RNN with other existing methods like

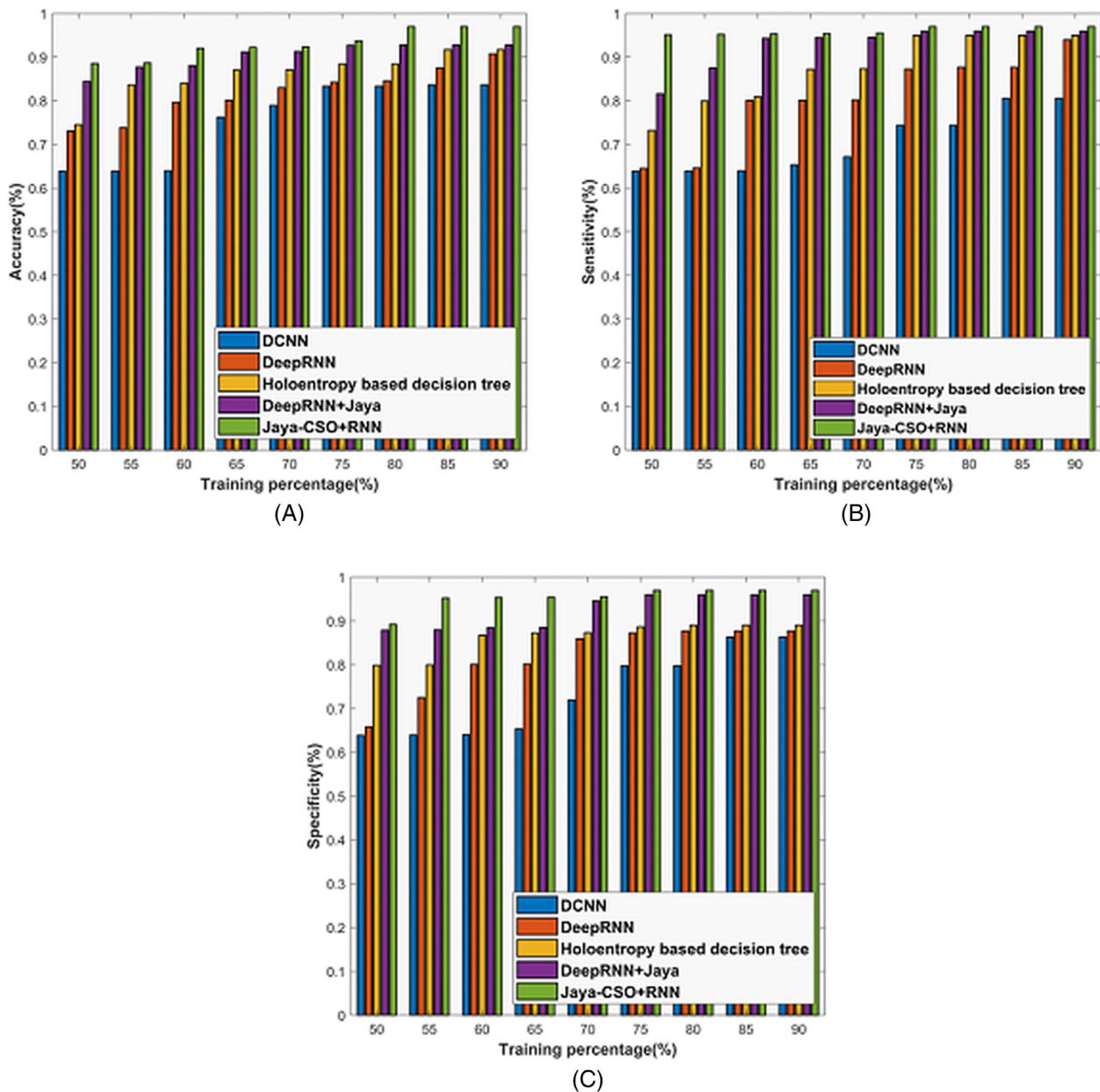


FIGURE 6 Analysis of proposed Jaya-CSO with image size 512 using (A) accuracy, (B) sensitivity, and (C) specificity [Color figure can be viewed at wileyonlinelibrary.com]

DCNN, DeepRNN, Holoentropy-based decision tree, DeepRNN + Jaya using accuracy, sensitivity, and specificity parameter.

The analysis is done using the image size 256 and 512. The maximum accuracy, sensitivity, and specificity are attained with image size =512 using 90% training data. Here, the maximal accuracy is computed by proposed Jaya-CSO + RNN with the highest accuracy value of 0.97, whereas the accuracy values computed by existing DCNN, DeepRNN, Holoentropy-based decision tree, DeepRNN + Jaya are 0.837, 0.907, 0.917, and 0.928.

Likewise, the maximal sensitivity is computed by proposed Jaya-CSO + RNN with the highest sensitivity value

of 0.97, whereas the sensitivity values computed by existing DCNN, DeepRNN, Holoentropy-based decision tree, DeepRNN + Jaya are 0.806, 0.94, 0.95, and 0.96. Similarly, the maximal specificity is computed by proposed Jaya-CSO + RNN with the highest specificity value of 0.97, whereas the specificity values computed by existing DCNN, DeepRNN, Holoentropy-based decision tree, DeepRNN + Jaya are 0.863, 0.877, 0.890, and 0.96.

The accuracy of the proposed Jaya-CSO + RNN is 13.71%, 6.49%, 5.46%, and 4.32% better than the existing methods, such as DCNN, DeepRNN, Holoentropy-based decision tree, and DeepRNN + Jaya, respectively, by using the image size 512. At image size 512, the

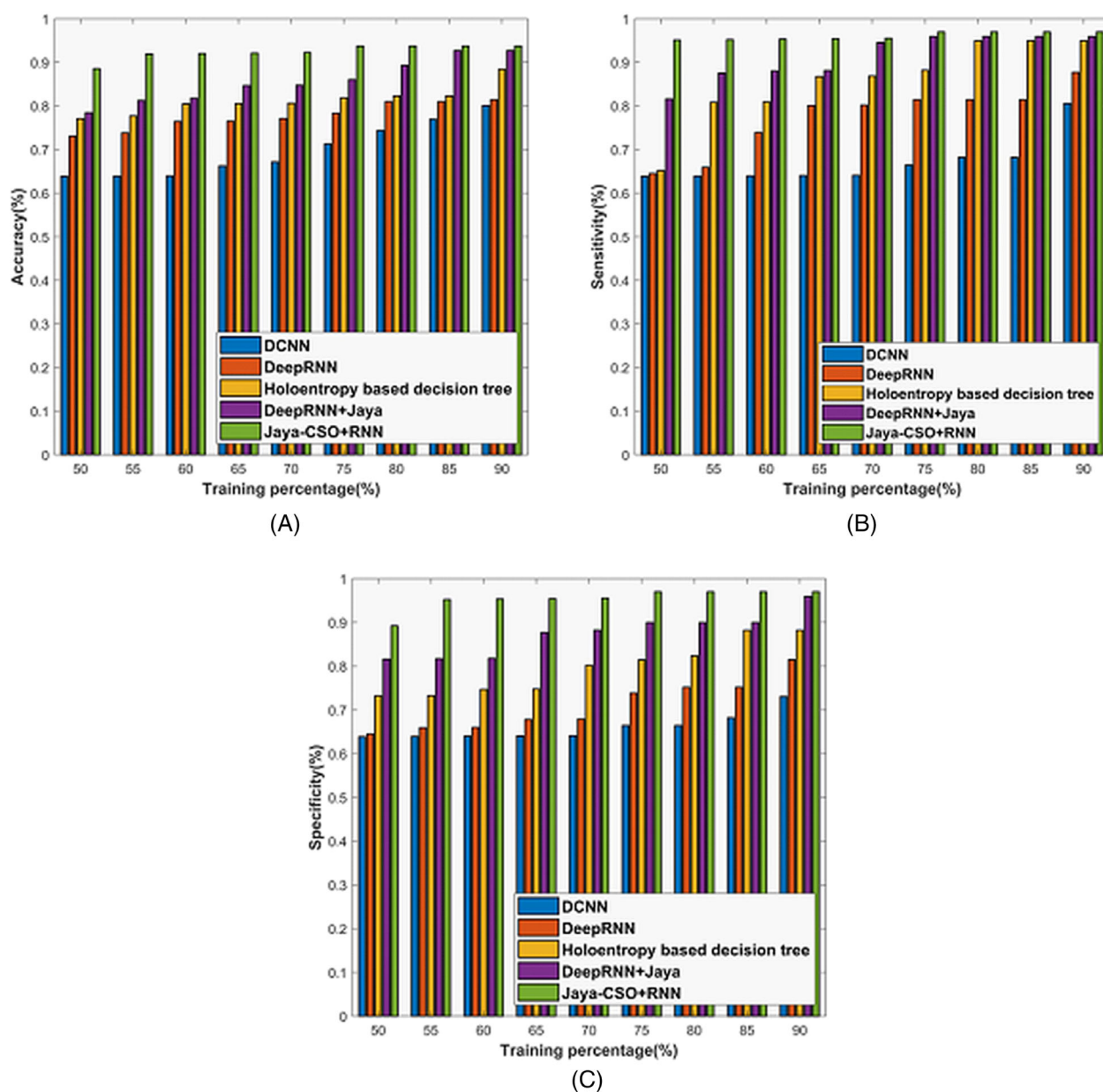


FIGURE 7 Analysis of proposed Jaya-CSO with image size 256 using (A) accuracy, (B) sensitivity, and (C) specificity [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/jna.22435)]

TABLE 1 Comparative analysis

Image size	Metrics	DCNN	DeepRNN	Holoentropy-based decision tree	DeepRNN + Jaya	Proposed Jaya-CSO + RNN
Using image size 512	Accuracy	0.837	0.907	0.917	0.928	0.97
	Sensitivity	0.806	0.94	0.95	0.96	0.97
	Specificity	0.863	0.877	0.890	0.96	0.97
Using image size 256	Accuracy	0.801	0.814	0.884	0.926	0.937
	Sensitivity	0.806	0.877	0.95	0.96	0.97
	Specificity	0.730	0.814	0.882	0.96	0.97

sensitivity of the proposed Jaya-CSO + RNN is 16.90%, 3.09%, 2.06%, and 1.03 better than the existing method such as DCNN, DeepRNN, Holoentropy-based decision tree, and DeepRNN + Jaya, respectively. Similarly, at image size 512, the proposed Jaya-CSO + RNN is 11.03%, 9.58%, 8.24%, and 1.03% better than the existing methods, such as DCNN, DeepRNN, Holoentropy-based decision tree, and DeepRNN + Jaya, respectively, based on the specificity.

The accuracy of the proposed Jaya-CSO + RNN is 14.51%, 13.12%, 5.65%, and 1.17% better than the existing methods, such as DCNN, DeepRNN, Holoentropy-based decision tree, and DeepRNN + Jaya, respectively, by using the image size 256. At image size 256, the proposed Jaya-CSO + RNN is 16.90%, 9.58%, 2.06%, and 1.03% better than the existing method such as DCNN, DeepRNN, Holoentropy-based decision tree, and DeepRNN + Jaya, respectively, based on the sensitivity. Similarly, at image size 256, the specificity of the proposed Jaya-CSO + RNN is 24.74%, 16.08%, 9.07%, and 1.03% better than the existing methods, such as DCNN, DeepRNN, Holoentropy-based decision tree, and DeepRNN + Jaya, respectively.

Table 2 depicts the comparative analysis of the proposed Jaya-CSO + RNN with other existing methods, which are present in the literature survey. The results shown are based on the parameters, such as accuracy, sensitivity, and specificity, without considering the dataset.

The accuracy of the proposed Jaya-CSO + RNN is 0.56%, 13.84%, 18.35%, 3.09%, 5.15%, 9.07%, 9.22%, and 17.52% better than the existing methods, such as HDT, LS-SVM, Centroid based thresholding, HTF, CDR + Feature-based detection, CNN + SVM classifier, VGG19 TL, and HRF, respectively.

TABLE 2 Comparative discussion

Method	Sensitivity (%)	Specificity (%)	Accuracy (%)
HDT ⁵	96.72	97.01	96.45
LS-SVM ⁸	86.4	80.80	83.57
Centroid based thresholding ¹⁰	87.41	72.19	79.20
HTF ¹¹	96	92	94
CDR + Feature based detection ²⁶	100	88	92
CNN + SVM classifier ³¹	85	90.80	88.20
VGG19 TL ³²	87.01	89.01	88.05
HRF ³³	83.33	77.78	80.00
Proposed Jaya-CSO + RNN	97	97	97

5 | CONCLUSIONS

A novel optimization-driven technique named Jaya-CSO + RNN is proposed for Glaucoma detection. The proposed Jaya-CSO is developed by combining the update equations of the Jaya algorithm and CSO algorithm for training the RNN classifier to provide optimal weights. At first, the input image is subjected to preprocessing for removing noise and artifacts present in the image. The obtained preprocessed image is fed to both optic disc detection and blood vessel detection.

In optic disc detection, the binarization and circle fitting technique is applied for extracting the significant features and simultaneously in blood vessel detection, the Renyi entropy-based sparking process is applied for extracting the essential features employed in the classification. The image-level features, optic disc level features, and blood vessel features are obtained from the process. Once the features are extracted, then the feature vector is established using available features. The feature vector is subjected to the RNN classifier, wherein the proposed Jaya-CSO is employed for training the RNN for glaucoma detection. The proposed method is effective in determining the glaucoma present in the input image. The proposed Jaya-CSO outperformed other existing with maximal accuracy of 0.97, the specificity of 0.97, and sensitivity of 0.97, respectively. The proposed method is only capable of detecting the glaucoma with maximal accuracy, and in the future, it will be further improved with some other algorithm to monitor the glaucoma progression.

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